

Cancel & Replace Classes

Handle order cancellation and modification requests

Typical flow: Cancel old order → Create new order → add to book

Must happen as one locked operation → Prevents another thread from seeing the book in a half-updated state.

{Struct} CancelRequest

+ UInt64_t orderId
+ std::string Symbol

Data Transfer Object

{Struct} ReplaceRequest

+ UInt64_t orderId
+ std::string Symbol
+ double newPrice (Optional implementation) std::optional<double>
+ double newQty (Optional implementation) std::optional<double>

Only fields being changed are sent

Data Transfer Object

Matching Engine

OrderFactory

Validates order exists
deletes it from engine

Depends on both
Matching Engine
and
OrderFactory

Cancel Processor

- std::shared_ptr<MatchEngine> matchEngine

+ bool processCancel(const CancelRequest & request)
- bool validateRequest(const CancelRequest & request)

Replace Processor

- std::shared_ptr<MatchEngine> matchEngine
- std::shared_ptr<OrderFactory> orderFactory

+ processReplacement(const ReplaceRequest & request)
- validateReplacement(const ReplaceRequest & request)